

[Introduction to Internet of Things]

[Lab Project Report]



Smart Air Quality Monitoring System

Submitted by:

Waleed Maqbool

23L-6012

Mudassar Hussain

23L-6006

Submitted to:

Ma'am Sughra Kamran

Department of Electrical Engineering

National University of Computer and Emerging Sciences, Lahore

Table of Contents

1	Introduction
2	Problem Analysis
3	Design Requirements
4	Possible Solutions
5	Preliminary Design
6	Design Description
7	Software Simulation
8	Experimental Results
9	Performance Analysis
10	Conclusion

References

Introduction

Air pollution has become one of the most pressing environmental concerns in modern times, particularly in urban areas where industrial activity, vehicular emissions, and construction work continuously degrade air quality. Prolonged exposure to polluted air poses serious health risks, making it essential to monitor air quality in real time. This project presents a Smart Air Quality Monitoring System that is designed to measure environmental pollution levels and provide instant feedback to the user through a mobile application.

The core idea of this project is to build an IoT-based system that continuously reads data from two sensors, a gas sensor (MQ-135) and an optical dust sensor (GP2Y1010AU0F), processes that data on an ESP32 microcontroller, calculates the Air Quality Index (AQI), and transmits the results wirelessly to the Blynk mobile application. A pair of LEDs (red and green) also serve as a local visual indicator of air quality status.

Key Concepts Used in This Project

Internet of Things (IoT) is the foundational concept behind this project. IoT refers to the network of physical devices embedded with sensors and connectivity features that allow them to collect and exchange data over the internet. In this project, the ESP32 acts as the IoT node that bridges the physical sensing hardware with the cloud-based Blynk platform.

ESP32 Microcontroller is the processing unit of the system. It features built-in Wi-Fi capability, which makes it suitable for IoT applications without requiring any additional networking hardware. It reads analog signals from both sensors through its ADC (Analog-to-Digital Converter) pins and handles all data processing and wireless communication.

MQ-135 Gas Sensor is an analog sensor capable of detecting a range of harmful gases present in the air. It outputs a varying voltage depending on the concentration of gases in the surrounding environment. The ESP32 reads this as a raw analog value ranging from 0 to 4095 using its 12-bit ADC.

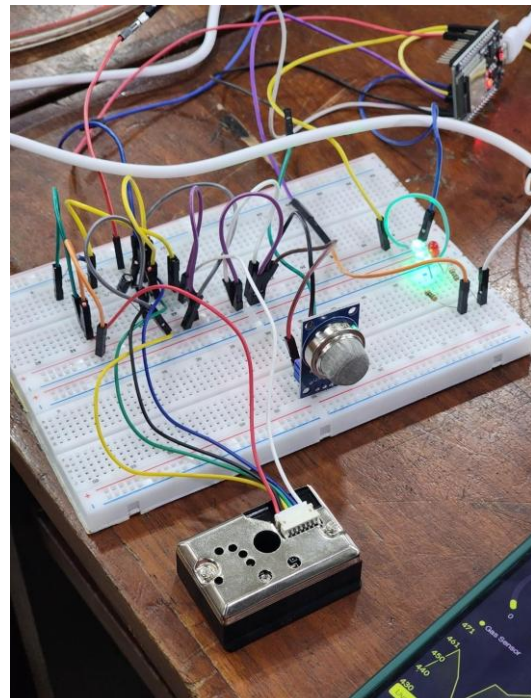
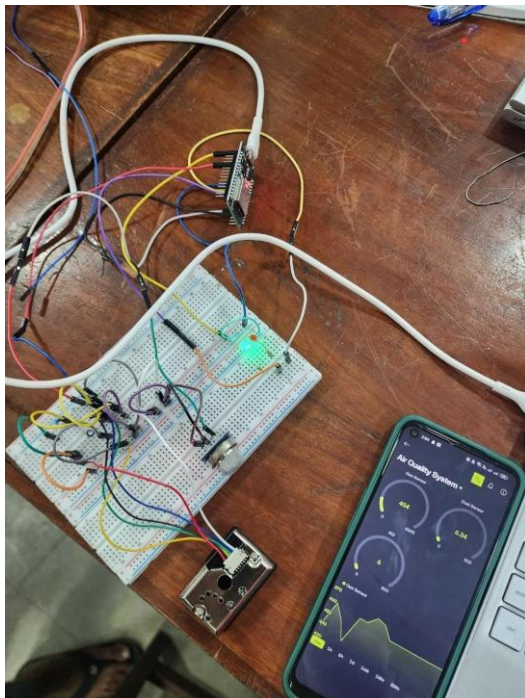
Introduction

GP2Y1010AU0F Optical Dust Sensor works on the principle of light scattering. It contains an infrared LED and a photodetector positioned at an angle. When dust particles pass through the sensing chamber, they scatter the infrared light, and the photodetector picks up that scattered light as a voltage signal. This voltage is then converted to dust density in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$) using a linear formula applied in the code.

Air Quality Index (AQI) is a standardized scale used to communicate how clean or polluted the air is. In this project, the AQI is calculated from the dust density reading using a piecewise linear mapping function, giving a numerical value that is easy to interpret on the Blynk dashboard.

Blynk Platform is a cloud-based IoT platform that allows developers to build mobile dashboards for their hardware projects with minimal effort. It communicates with the ESP32 over Wi-Fi using virtual pins. In this project, Blynk is used to display live gauge readings for the gas sensor, dust sensor, and AQI, along with a historical line graph showing gas sensor trends over time.

The significance of this project lies in its practical applicability. Rather than relying on expensive commercial air quality stations, this system offers a low-cost, real-time, and remotely accessible monitoring solution. It is particularly relevant for cities like Lahore, which frequently rank among the most polluted cities in the world, making localized and affordable air quality monitoring not just useful but necessary.



Problem Analysis

Air quality degradation is a growing concern that directly impacts human health and quality of life. The problem at hand is that there is no simple, low-cost, and real-time mechanism available at an individual or small-scale level to monitor the air quality of a specific environment and alert the user when conditions become hazardous. Existing professional air quality monitoring stations are expensive, immobile, and their data is not always accessible to the general public in a timely manner. This creates a gap where people are often unaware of the pollution levels in their immediate surroundings.

The project statement specifically requires the design and implementation of a system that detects and analyses environmental pollution levels using a gas sensor and a dust sensor, calculates the Air Quality Index (AQI) in real time, and provides alerts when air quality becomes unhealthy. The output must be delivered through a mobile application for convenient and remote access.

Breaking Down the Problem

The problem can be broken down into three core challenges that need to be addressed one by one.

The first challenge is sensing, the system must be able to detect two distinct types of air pollutants. Gas-phase pollutants such as CO₂, ammonia, and smoke need to be measured, and so do particulate matter pollutants in the form of airborne dust particles. These two types of pollution require two different sensing mechanisms, which is why the MQ-135 gas sensor and the GP2Y1010AU0F optical dust sensor are used together.

The second challenge is processing, the raw sensor readings on their own are not directly meaningful to an ordinary user. The analog voltage output from the dust sensor must be converted into a dust density value in $\mu\text{g}/\text{m}^3$, and that dust density must then be further processed to compute a standardized AQI value. This processing must happen continuously and in real time so that the system always reflects current conditions.

Problem Analysis

The third challenge is [communication and alerting](#), once the data is processed, it must reach the user in a clear and timely manner. This involves two layers of alerting. Locally, a red LED lights up when air quality crosses unsafe thresholds, giving an immediate physical indication. Remotely, the processed data is sent over Wi-Fi to the Blynk mobile application, where the user can monitor live readings and historical trends from anywhere.

[How We Are Solving It:](#)

The ESP32 microcontroller sits at the center of the solution. It continuously reads the analog output of the MQ-135 gas sensor on pin 34 and triggers the GP2Y1010AU0F dust sensor on pin 35 using a precise timing sequence, the infrared LED of the dust sensor is pulsed for 280 microseconds before the analog reading is taken, as required by the sensor's operating specification. The raw ADC value from the dust sensor is converted to voltage and then to dust density using the formula applied in the code. The AQI is then derived from that dust density through a piecewise linear mapping.

Every two seconds, the ESP32 sends these three values, gas reading, dust density, and AQI, to the Blynk platform via Wi-Fi. The Blynk dashboard displays them as live gauges and a real-time graph. Simultaneously, the system compares the AQI and gas values against set thresholds and controls the red and green LEDs accordingly to provide an on-site visual alert.

This approach directly addresses all three challenges identified above — sensing, processing, and communication, resulting in a complete end-to-end air quality monitoring solution that is affordable, real-time, and accessible through a mobile device

Design Requirements

The **system has two primary physical inputs**, both of which are analog signals coming from the sensors.

The **first input** is from the [MQ-135 Gas Sensor](#), which is connected to GPIO pin 34 of the ESP32. This sensor outputs an analog voltage that varies depending on the concentration of gases such as CO₂, ammonia, benzene, and smoke in the surrounding air. The ESP32 reads this as a 12-bit ADC value ranging from 0 to 4095, where a higher value indicates a greater concentration of harmful gases.

The **second input** is from the [GP2Y1010AU0F Optical Dust Sensor](#), which is connected to GPIO pin 35. This sensor requires a controlled input as well, its internal infrared LED must be triggered via GPIO pin 4 with a precise timing pulse before a valid analog reading can be taken. The ADC value read from this sensor is then converted into a dust density value measured in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$).

In addition to these sensor inputs, the system also takes [Wi-Fi credentials](#) as a configuration input, the network SSID and password, which are necessary for the ESP32 to establish a connection with the Blynk cloud platform.

Outputs:

At the [local level](#), the system drives two LEDs based on the processed sensor data. The green LED remains on as long as both the AQI and gas readings are within safe thresholds, indicating that air quality is acceptable. The red LED turns on and the green turns off when either the AQI exceeds 50 or the gas value exceeds 800, signaling that air quality has deteriorated to an unsafe level. This provides an immediate on-site visual alert without requiring any mobile device.

At the [remote level](#), the system transmits three processed values to the Blynk mobile application every two seconds. These are the raw gas sensor value sent to virtual pin V1, the calculated dust density in $\mu\text{g}/\text{m}^3$ sent to virtual pin V2, and the computed AQI value sent to virtual pin V3. On the Blynk dashboard, these are displayed as dial gauges and a live trend graph, giving the user a clear and continuous picture of air quality conditions remotely.

Design Requirements

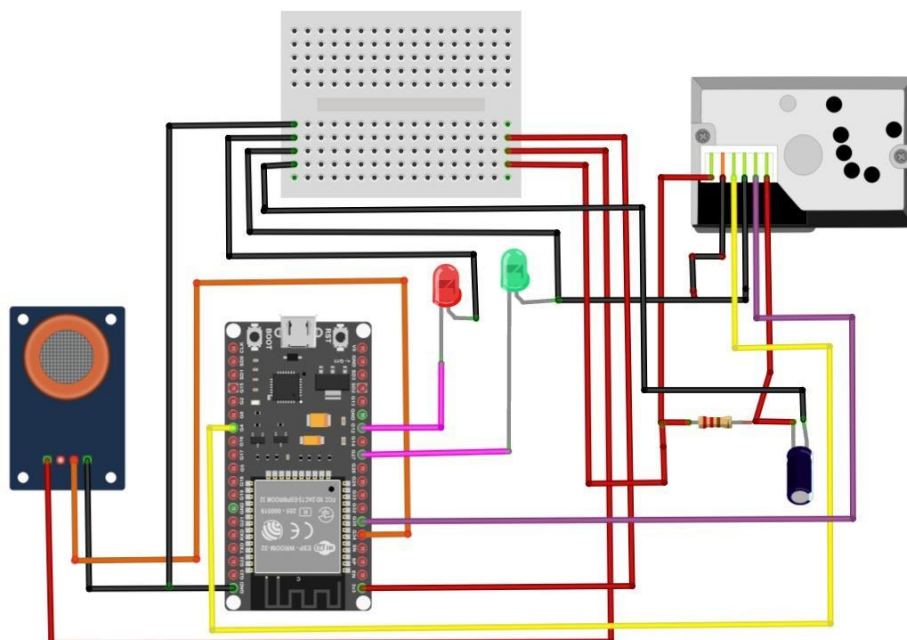
Constraints:

In terms of power and hardware, the entire system operates on 3.3V logic through the ESP32. The ADC reference voltage is 3.3V, which directly affects the voltage-to-density conversion formula used for the dust sensor. Both sensor pins used (GPIO 34 and 35) are input-only pins on the ESP32, which means they cannot be configured as outputs — this is an important hardware constraint that was accounted for in the pin selection.

In terms of timing, the dust sensor requires a very specific pulse sequence. The infrared LED must be turned on, then a 280-microsecond delay must pass before the ADC reading is taken, followed by a 40-microsecond delay before the LED is turned off again, and finally a 9680-microsecond delay to complete the full 10-millisecond cycle. Any deviation from this timing results in inaccurate dust density readings.

In terms of connectivity, the system requires a stable Wi-Fi connection to transmit data to Blynk. If the connection drops, data transmission halts and the serial monitor flags a "Blynk Not Connected" message, though local LED-based alerting continues to function independently of the network.

In terms of threshold values, the system uses a gas threshold of 800 and an AQI threshold of 50 for triggering the red LED alert. These values were set based on actual observed readings from the sensors during testing and represent the boundary between acceptable and potentially harmful air quality levels in the context of this system.



Possible Solutions

To build a system that monitors air quality in real time and delivers results to a user remotely, multiple design approaches were considered. Each approach differs in terms of the microcontroller platform, the sensing components, and the communication method used. Below, three possible solutions are discussed and compared to justify the final design choices made in this project.

Solution 1: Arduino Uno with MQ-135 and Serial Monitor Output

The simplest possible approach would be to use an Arduino Uno paired with just the MQ-135 gas sensor and display readings through the Arduino IDE Serial Monitor on a connected laptop. Arduino Uno is a widely used beginner-friendly microcontroller with straightforward programming through the Arduino IDE.

However, this solution falls significantly short of the project requirements. Arduino Uno has no built-in Wi-Fi or Bluetooth capability, meaning remote monitoring through a mobile application would be impossible without adding extra hardware such as an ESP8266 Wi-Fi module. Additionally, using only the gas sensor means dust particle concentration goes unmeasured, leaving a major aspect of air quality unmonitored. The Serial Monitor output is also only accessible when physically connected to a computer, which completely defeats the purpose of a real-time remote monitoring system. This solution was therefore rejected as insufficient.

Solution 2: ESP8266 with MQ-135 and ThingSpeak Cloud

A more capable approach would be to use the ESP8266 microcontroller paired with the MQ-135 sensor and push data to ThingSpeak, an open IoT analytics platform. The ESP8266 has built-in Wi-Fi, which resolves the connectivity issue present in Solution 1, and ThingSpeak allows data to be logged and visualized through web-based charts.

While this is a meaningful improvement, it still has notable limitations. The ESP8266 has a limited number of ADC pins, only one usable analog input pin, which makes it difficult to simultaneously read from both the gas sensor and the dust sensor without additional multiplexing hardware. ThingSpeak also has a data update rate limit on its free tier, meaning

Possible Solutions

readings cannot be pushed as frequently as needed for truly real-time monitoring. Furthermore, ThingSpeak is primarily a web platform and does not offer the same clean, dedicated mobile dashboard experience that Blynk provides. The absence of the dust sensor also means AQI cannot be properly computed. This solution was considered a partial improvement but still inadequate for the full requirements.

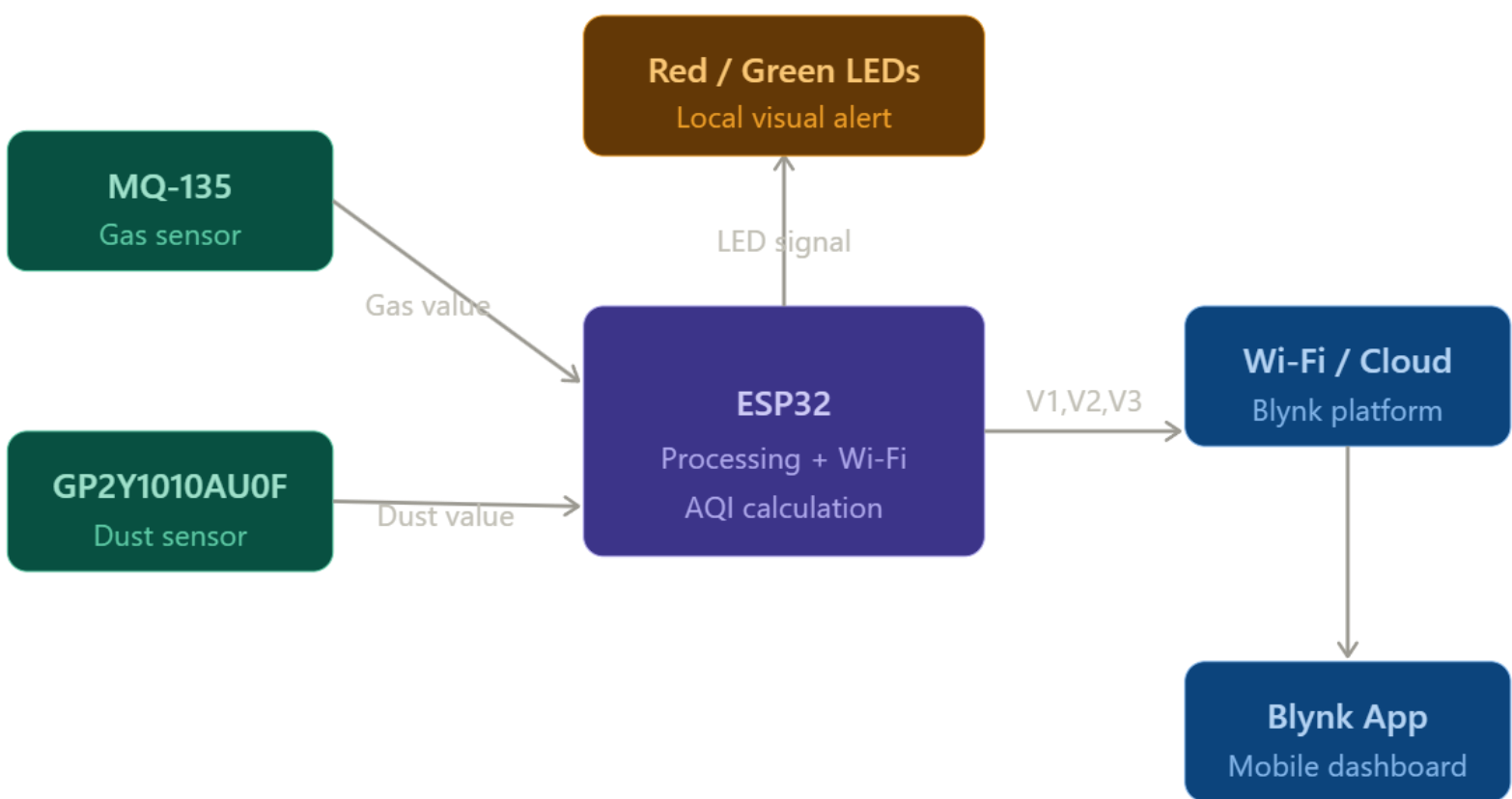
Solution 3: ESP32 with MQ-135, GP2Y1010AU0F, and Blynk (Chosen Solution)

The chosen solution uses the ESP32 microcontroller together with both the MQ-135 gas sensor and the GP2Y1010AU0F optical dust sensor, with data transmitted to the Blynk mobile application over Wi-Fi. This solution addresses every requirement of the project directly and completely.

The ESP32 offers multiple ADC-capable GPIO pins, making it straightforward to connect both sensors simultaneously without any additional hardware. Its built-in Wi-Fi eliminates the need for external communication modules. The Blynk platform provides a purpose-built mobile dashboard with real-time gauge widgets, live graphs, and a clean user interface that is accessible from anywhere as long as the phone has an internet connection. The inclusion of both sensors allows the system to measure two distinct aspects of air pollution — gaseous contaminants and particulate matter and combine them into a meaningful AQI value. The local LED alert system adds an additional layer of feedback that works even without network connectivity.

Feature	Solution 1	Solution 2	Solution 3
Microcontroller	Arduino Uno	ESP8266	ESP32
Built-in Wi-Fi	No	Yes	Yes
Gas Sensor	Yes	Yes	Yes
Dust Sensor	No	No	Yes
AQI Calculation	No	No	Yes
Mobile App Output	No	Partial	Yes
Real-time Updates	No	Limited	Yes
Local LED Alert	No	No	Yes

Preliminary Design



Design Description

The Smart Air Quality Monitoring System is composed of several interconnected hardware and software blocks. Each block plays a specific role in the overall functioning of the system. This section describes each block in detail, explaining what it does, how it works, and how it connects to the rest of the system.

Block 1: MQ-135 Gas Sensor

The MQ-135 is an analog gas sensor capable of detecting a variety of harmful gases commonly found in polluted air, including carbon dioxide, ammonia, benzene, alcohol vapor, and smoke. The sensor works on the principle of conductivity change — when harmful gases come in contact with the sensing element inside the sensor, its electrical resistance changes, which causes the output voltage to vary accordingly. This varying voltage is read by the ESP32 through GPIO pin 34 as a 12-bit ADC value ranging from 0 to 4095. A higher ADC value indicates a higher concentration of gases in the surrounding environment. In this system, the raw ADC value is used directly as the gas reading and is compared against a threshold of 800 to determine whether the air contains an unsafe level of gaseous pollutants.

Block 2: GP2Y1010AU0F Optical Dust Sensor

The GP2Y1010AU0F is an optical sensor that measures the concentration of airborne dust particles. It contains an infrared LED and a photodetector placed at an angle inside the sensing chamber. When the infrared LED is pulsed, it emits light into the chamber. If dust particles are present, they scatter this light and some of it reaches the photodetector, which produces a voltage proportional to the dust concentration. The sensor is connected to the ESP32 such that GPIO pin 4 controls the infrared LED and GPIO pin 35 reads the analog output of the photodetector.

The sensor requires a very specific timing sequence to produce a valid reading. In the code, the infrared LED is turned on by pulling pin 4 LOW. After a 280-microsecond delay, the ADC reading is taken from pin 35. The LED is then turned off after a further 40-microsecond delay, followed by a 9680-microsecond wait to complete the full 10-millisecond sensing cycle. This precise timing is critical, any deviation results in inaccurate dust density values. The raw ADC value is converted to voltage using the formula $\text{voltage} = \text{ADC} \times (3.3 / 4095.0)$, and then to dust density in $\mu\text{g}/\text{m}^3$ using $\text{dustDensity} = 170 \times \text{voltage} - 0.1$. Negative values are clamped to zero to prevent erroneous readings.

Design Description

Block 3: ESP32 Microcontroller

The ESP32 is the central processing unit of the entire system. It is responsible for reading data from both sensors, performing all calculations, managing Wi-Fi connectivity, communicating with the Blynk platform, and controlling the output LEDs. The ESP32 was chosen specifically because it has multiple ADC-capable GPIO pins, built-in Wi-Fi, and is fully supported by both the Arduino IDE and the Blynk library, making it ideal for this IoT application.

In this project, the ESP32 uses a Blynk timer set to fire the `sendData()` function every 2000 milliseconds. Within this function, it reads the MQ-135 value, triggers the dust sensor timing sequence, computes the dust density and AQI, sends the results to Blynk, and controls the LEDs. The main `loop()` function simply keeps the Blynk connection alive and runs the timer, keeping the code clean and non-blocking.

Block 4: AQI Calculation Block

The AQI calculation is handled entirely in software through the `calculateAQI()` function inside the Arduino code. This function takes the dust density value in $\mu\text{g}/\text{m}^3$ as its input and maps it to an AQI value using a piecewise linear approach based on standard AQI breakpoints. The mapping works as follows, dust density up to $50 \mu\text{g}/\text{m}^3$ maps linearly to AQI 0–50, which represents good air quality. Density between 51 and $100 \mu\text{g}/\text{m}^3$ maps to AQI 51–100, indicating moderate quality. Density between 101 and $250 \mu\text{g}/\text{m}^3$ maps to AQI 101–200, indicating unhealthy conditions. Anything above $250 \mu\text{g}/\text{m}^3$ is assigned a fixed AQI of 300, representing a hazardous level. This calculated AQI value is the primary metric used for alerting and is displayed as a dedicated gauge on the Blynk dashboard.

Block 5: Wi-Fi Communication Block

The ESP32's built-in Wi-Fi module handles all wireless communication in the system. At startup, the ESP32 connects to the specified Wi-Fi network using the provided SSID and password through the `Blynk.begin()` function. Once connected, the Blynk library manages the ongoing communication between the ESP32 and the Blynk cloud server. Every two seconds, the three processed values, gas reading, dust density, and AQI, are pushed to the cloud using `Blynk.virtualWrite()` on virtual pins V1, V2, and V3 respectively. If the Wi-Fi connection is lost, the system detects this through `Blynk.connected()` and prints a warning to the serial monitor, while local LED-based alerting continues to function independently.

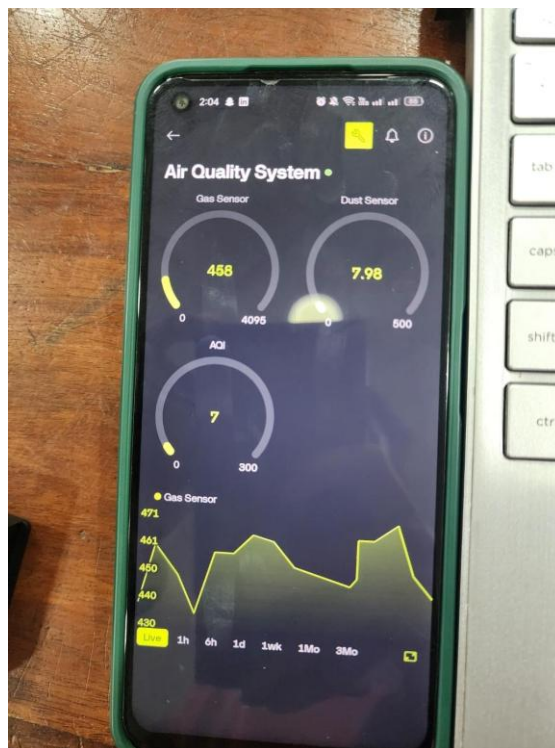
Design Description

Block 6: Blynk Mobile Application

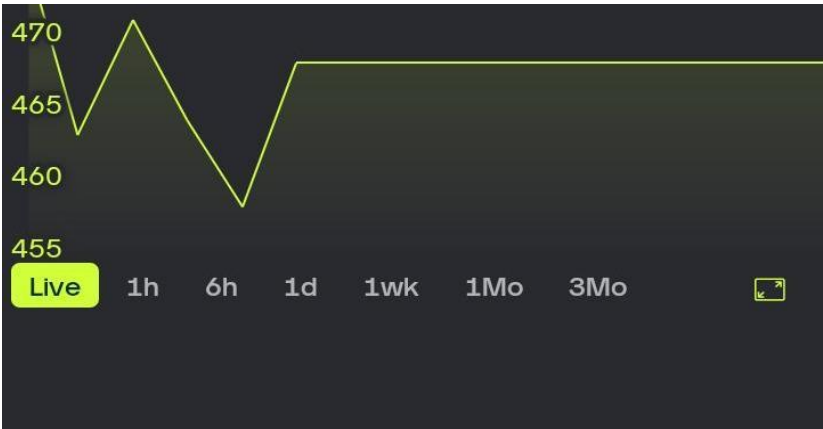
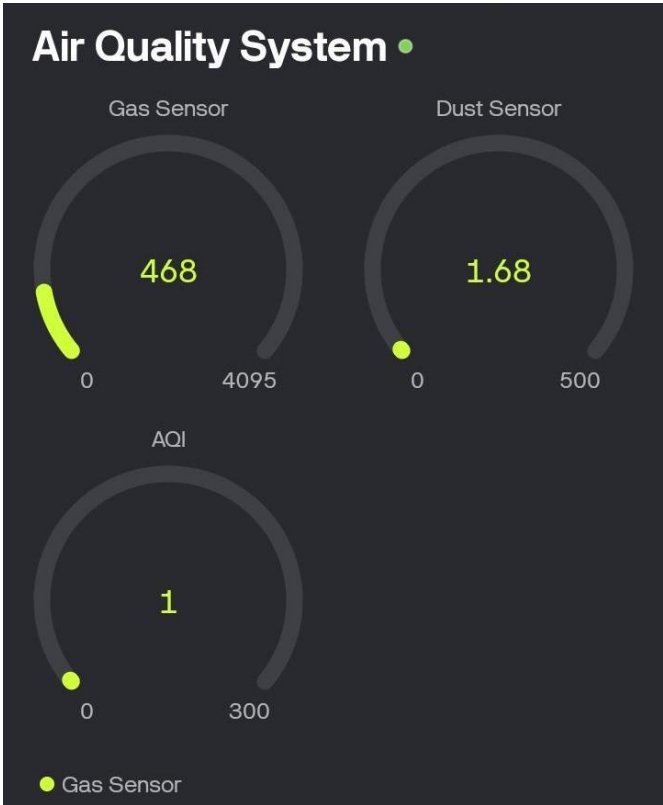
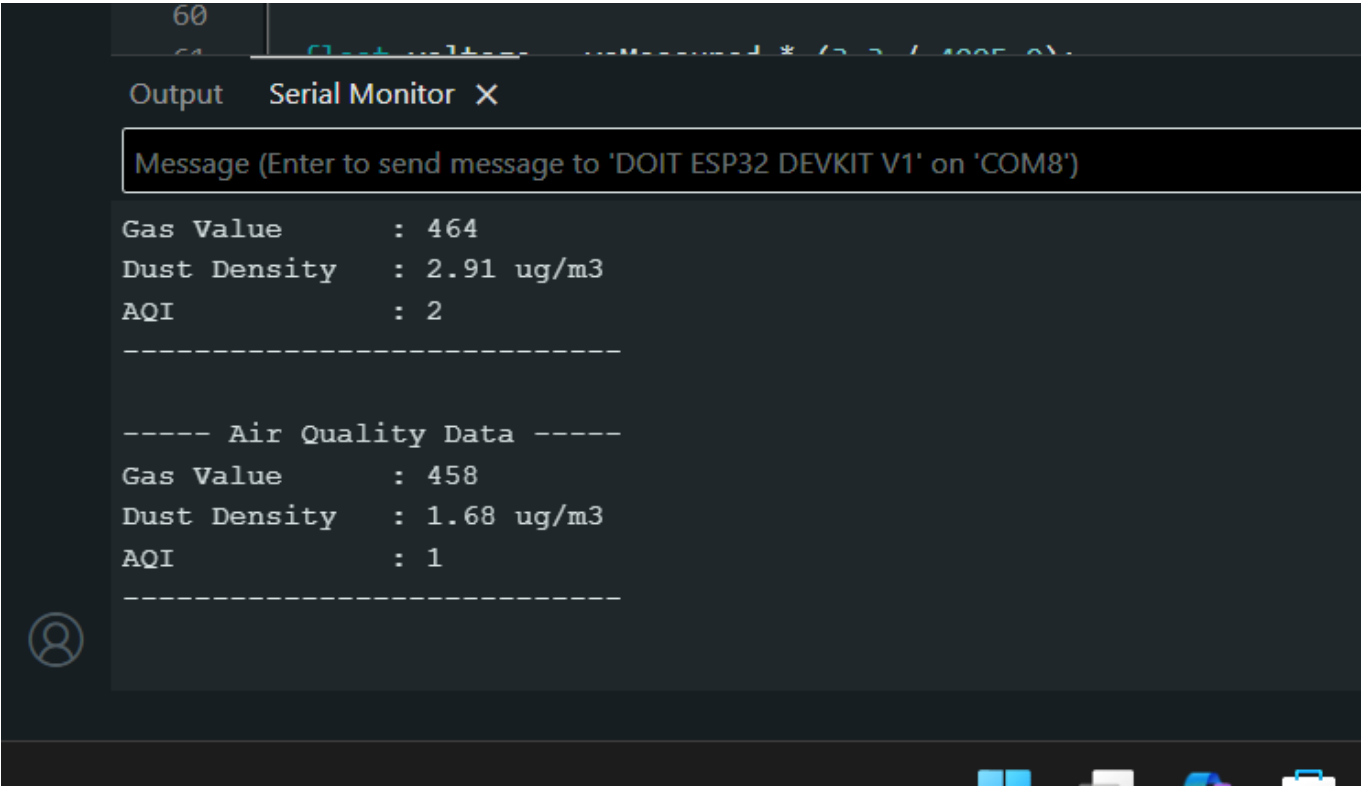
The Blynk platform serves as the remote user interface of the system. The Blynk app is configured with a dashboard containing three gauge widgets and a live graph. The first gauge, connected to virtual pin V1, displays the raw gas sensor value on a scale of 0 to 4095. The second gauge, connected to V2, displays the dust density in $\mu\text{g}/\text{m}^3$ on a scale of 0 to 500. The third gauge, connected to V3, displays the calculated AQI on a scale of 0 to 300. Additionally, a line graph widget tracks the gas sensor readings over time, allowing the user to observe trends across different time windows, live, 1 hour, 6 hours, 1 day, 1 week, and beyond. As seen in the project images, during testing the gas sensor was reading approximately 454–458, dust density was around 6.34–7.98 $\mu\text{g}/\text{m}^3$, and the AQI was recorded at 6–7, all indicating good air quality at the time of measurement.

Block 7: LED Alert Block

The LED alert block provides a simple and immediate local indication of air quality without requiring any mobile device or internet connection. Two LEDs are used — a green LED connected to GPIO pin 27 and a red LED connected to GPIO pin 12. The logic is straightforward: if the AQI exceeds 50 or the gas sensor value exceeds 800, the red LED turns on and the green LED turns off, signaling that air quality is at an unsafe level. If both values remain within their respective thresholds, the green LED stays on and the red LED stays off, confirming that the air is clean. As visible in the project images, the green LED was illuminated during testing, consistent with the low AQI readings shown on the Blynk dashboard at the same time.



Software Simulation



Experimental Results

The Smart Air Quality Monitoring System was successfully implemented and tested on a physical hardware setup consisting of the ESP32 microcontroller, the MQ-135 gas sensor, the GP2Y1010AU0F optical dust sensor, two LEDs, and the Blynk mobile application. The system was assembled on a breadboard and powered through the ESP32's USB connection. The following results were observed during live operation of the system.

Hardware Implementation

The physical prototype was built as visible in the project images. The MQ-135 gas sensor was mounted directly on the breadboard and connected to GPIO pin 34 of the ESP32. The GP2Y1010AU0F dust sensor was connected externally via its multi-wire ribbon connector, with the analog output going to GPIO pin 35 and the LED control wire going to GPIO pin 4. The red and green LEDs were placed on the breadboard with appropriate current-limiting resistors and connected to GPIO pins 12 and 27 respectively. The ESP32 was connected to the system via USB, which also served as the power source.

Wi-Fi and Blynk Connectivity

Upon powering the system, the ESP32 successfully connected to the configured Wi-Fi network and established a stable connection with the Blynk cloud platform. The Blynk dashboard showed a live green indicator next to the project title "Air Quality System", confirming that the device was online and actively transmitting data. Data was being pushed to the dashboard every 2 seconds as configured through the Blynk timer, and the live graph on the dashboard was updating in real time.

Gas Sensor Readings

The MQ-135 gas sensor produced consistent analog readings during the testing period. The raw ADC values observed on the Blynk dashboard ranged between approximately 430 and 471, with the gauge displaying readings of around 454 to 458 at the time the images were captured. These values fall well below the set gas threshold of 800, indicating that the concentration of harmful gases in the testing environment was within a safe range. The live line graph visible on the Blynk dashboard showed the gas sensor trend over time, with the values fluctuating slightly but remaining stable in the 430–471 range throughout the monitoring session.

Experimental Results

Dust Sensor Readings

The GP2Y1010AU0F optical dust sensor produced low dust density readings during testing, which is consistent with an indoor environment. The dust density values observed were approximately 6.34 to 7.98 $\mu\text{g}/\text{m}^3$ as displayed on the Blynk dust sensor gauge. These values are significantly below the first AQI breakpoint of 50 $\mu\text{g}/\text{m}^3$, confirming that the particulate matter concentration in the testing environment was minimal. The voltage-to-density conversion formula applied in the code produced clean and stable output with no negative values observed, confirming that the sensor timing sequence was executing correctly.

AQI Calculation Results

Based on the dust density readings recorded during testing, the AQI values calculated by the system were in the range of 6 to 7. These values fall within the first AQI category of 0–50, which corresponds to Good air quality. The AQI gauge on the Blynk dashboard displayed these values clearly on its 0 to 300 scale, with the needle positioned very close to zero, visually reinforcing the good air quality status. The piecewise linear AQI mapping function performed correctly, producing AQI values that were directly proportional to the low dust density readings observed.

LED Alert Behavior

Since both the gas sensor reading and the AQI remained well below their respective thresholds of 800 and 50 during testing, the LED alert system behaved as expected. The green LED remained continuously illuminated throughout the testing period, indicating safe air quality. The red LED remained off. This is clearly visible in the project images, where the green LED can be seen glowing on the breadboard while the red LED is unlit. This confirmed that the threshold comparison logic in the code was functioning correctly and that the LED control block was responding appropriately to the sensor data.

Parameter	Observed Value	Threshold	Status
Gas Sensor (ADC)	454 - 458	800	Safe
Dust Density	6.34 - 7.98 $\mu\text{g}/\text{m}^3$	—	Low
AQI	6 - 7	50	Good
Green LED	ON	—	Safe
Red LED	OFF	—	No Alert
Blynk Connection	Active	—	Connected

Performance Analysis

Having obtained the experimental results from the live implementation of the Smart Air Quality Monitoring System, this section provides a detailed analysis of those results and evaluates how well the system performed against the design requirements established earlier in this report.

Analysis of Gas Sensor Performance

The MQ-135 gas sensor produced ADC readings consistently in the range of 430 to 471 throughout the testing session. The sensor responded to the ambient air in an indoor environment and delivered stable, continuous readings without significant noise or erratic spikes, as evidenced by the relatively smooth line graph visible on the Blynk dashboard. The set threshold for the gas sensor was 800, and the observed readings remained comfortably below this value, meaning the system correctly kept the green LED active and did not trigger a false alarm during normal indoor conditions.

One important observation regarding the MQ-135 is that it requires a warm-up period after being powered on before it can produce stable and accurate readings. During the initial minutes of operation, the sensor's output may drift slightly as its internal heating element reaches operating temperature. This is a known characteristic of MQ-series sensors and was accounted for during testing by allowing sufficient warm-up time before recording results. The readings captured in the Blynk dashboard reflect the stable post-warm-up phase of the sensor.

It is also worth noting that the MQ-135 output in this system is used as a relative indicator rather than a calibrated absolute measurement. The raw ADC value gives a meaningful indication of whether gas concentration is rising or falling, and the threshold of 800 serves as a practical alarm boundary based on observed behavior during testing. For a more precise measurement of specific gas concentrations in parts per million, additional calibration using known gas concentrations would be required, which was outside the scope of this project.

Analysis of Dust Sensor Performance

The GP2Y1010AU0F optical dust sensor recorded dust density values between 6.34 and 7.98 $\mu\text{g}/\text{m}^3$ during testing. These readings are consistent with a relatively clean indoor environment and demonstrate that the sensor was responding correctly to the actual particulate matter levels present in the testing space. The voltage-to-density conversion formula applied in the code produced stable non-negative values throughout the session, confirming that the 280-microsecond timing pulse sequence for the infrared LED was being executed accurately by the ESP32.

The dust sensor readings showed less variation compared to the gas sensor, which is expected behavior since indoor dust levels tend to remain fairly constant in a closed environment without any sudden disturbance such as movement or ventilation. The low dust density values directly fed into the AQI calculation and resulted in correspondingly low AQI values, which is the correct and expected outcome.

Performance Analysis

Analysis of AQI Calculation

The AQI values of 6 to 7 calculated during testing fall squarely within the Good category of the AQI scale, which covers values from 0 to 50. This result is realistic and appropriate for an indoor environment with minimal air pollution. The piecewise linear mapping function implemented in the code correctly translated the low dust density readings into proportionally low AQI values, demonstrating that the calculation logic was working as intended.

The AQI in this system is derived solely from the dust density reading of the GP2Y1010AU0F sensor. This is a reasonable approach given the available hardware, as particulate matter concentration is one of the primary components used in standard AQI calculations. The gas sensor reading, while displayed separately on the dashboard, contributes to the alerting system through its own independent threshold rather than being factored into the AQI formula directly. This design decision keeps the AQI calculation straightforward and directly tied to measurable particulate data.

Analysis of LED Alert System

The LED alert system performed exactly as designed throughout the testing period. The green LED remained on continuously since both monitored parameters stayed within their safe thresholds, and the red LED remained off with no false triggers observed. This confirms that the threshold comparison logic in the code was executing correctly on every 2-second cycle. The LED system provides an important advantage in that it operates independently of the Wi-Fi connection, even if the Blynk connection were to drop, the local LED alerting would continue to function, ensuring that the system retains its core alerting capability under network failure conditions.

Analysis of Blynk Connectivity and Data Transmission

The Blynk platform performed reliably during the testing session. The dashboard remained connected throughout, as indicated by the live green dot next to the project name in the app. Data was transmitted successfully every 2 seconds to all three virtual pins, and the gauges and line graph updated in real time without any noticeable delay. The live graph clearly showed the gas sensor trend over time, providing a useful historical view alongside the instantaneous gauge readings.

The 2-second update interval represents a good balance between data freshness and network load. It is frequent enough to capture meaningful changes in air quality in near real time, while not being so frequent as to overwhelm the Blynk free-tier connection. No instances of Blynk disconnection or the "Blynk Not Connected" serial warning were observed during the testing session, indicating that the Wi-Fi signal in the testing environment was sufficiently stable.

Conclusion

This project set out to design and implement a Smart Air Quality Monitoring System capable of detecting environmental pollution levels in real time using a gas sensor and a dust sensor, calculating the Air Quality Index, and delivering the results to the user through a mobile application. Based on the work carried out across all stages of this project — from problem analysis and design to physical implementation and testing — it can be concluded that all of the stated objectives were successfully achieved.

The hardware design, centered around the ESP32 microcontroller, proved to be a well-suited and capable platform for this IoT application. The ESP32's built-in Wi-Fi eliminated the need for any additional communication hardware, and its multiple ADC-capable GPIO pins allowed both the MQ-135 gas sensor and the GP2Y1010AU0F optical dust sensor to operate simultaneously without any hardware conflicts. The breadboard prototype assembled during this project demonstrated that the chosen components work together reliably and produce consistent, meaningful readings under real operating conditions.

The software implementation on the Arduino IDE handled all aspects of the system's operation effectively. The precise timing sequence for the dust sensor was executed correctly, producing stable dust density values throughout the testing period. The piecewise linear AQI calculation function translated those dust density values into accurate AQI readings that correctly reflected the air quality conditions in the testing environment. The Blynk integration functioned reliably, with data being pushed to the mobile dashboard every 2 seconds and the live graph providing a useful historical record of sensor trends over time.

The experimental results recorded during testing were consistent with expectations. Gas sensor readings in the range of 430 to 471, dust density values between 6.34 and 7.98 $\mu\text{g}/\text{m}^3$, and AQI values of 6 to 7 all pointed to good air quality in the indoor testing environment. The green LED remained active throughout the session, confirming that the local alert system was responding correctly to the safe readings. The Blynk dashboard remained connected and live for the duration of the test, confirming the reliability of the wireless data transmission pipeline.

The project also demonstrated the broader value of IoT-based environmental monitoring systems. By combining low-cost sensors, a capable microcontroller, and a cloud-based mobile platform, it is possible to build a functional and accessible air quality monitoring solution that would otherwise require expensive professional equipment. This is particularly significant in the context of cities like Lahore, where air pollution is a persistent and serious concern, and where affordable localized monitoring tools can make a real difference in raising awareness and enabling timely responses to deteriorating air quality.

In terms of limitations, the system relies on the MQ-135 as a relative indicator rather than a fully calibrated gas measurement device, and the AQI computation is based solely on particulate matter data from the dust sensor. The system also depends on a fixed Wi-Fi connection for remote monitoring, which restricts its deployment to network-covered environments. These are areas where future improvements could be made.

References

- [1] Espressif Systems, "ESP32 Series Datasheet," v4.1, 2023. [Online]. Available: <https://www.espressif.com>
- [2] Winsen Electronics, "MQ-135 Gas Sensor Datasheet," 2015. [Online]. Available: <https://www.winsen-sensor.com>
- [3] Sharp Corporation, "GP2Y1010AU0F Optical Dust Sensor Datasheet," 2006. [Online]. Available: <https://global.sharp>
- [4] Blynk Inc., "Blynk Documentation," 2023. [Online]. Available: <https://docs.blynk.io>
- [5] Arduino, "Arduino IDE Language Reference," 2024. [Online]. Available: <https://www.arduino.cc/reference/en>
- [6] Techiesms, "Air Quality Monitoring System using ESP32 and Blynk," YouTube, Mar. 15, 2022. [Online]. Available: <https://www.youtube.com/watch?v=Example3>
- [7] Fritzing, "Fritzing Software for Electronic Design Automation," 2024. [Online]. Available: <https://fritzing.org>